

Operational Research II, Midterm 2, 2026 Spring

Solution and Grading

- Duration: 120 minutes
- Closed material, No calculator

- Name: _____
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- Write legibly.
- Justification is necessary unless stated otherwise.

1	20
2	20
3	15
4	15
Total	70

#1. Fill in the blanks by using the following terms. [Each 2pts]

periodic	aperiodic	class	recurrent	transient	absorbing
finite	infinite	irreducible	reducible	reach	communicate

(a)	recurrent
(b)	periodic
(c)	reducible
(d)	aperiodic
(e)	irreducible
(f)	class
(g)	finite
(h)	irreducible
(i)	aperiodic
(j)	recurrent

- An absorbing state is a special case of a [(a)] state.
- A DTMC is said to be [(b)] if there exists a state whose period is higher than one.
- A DTMC is said to be [(c)] if there are more than one class.
- If all diagonal elements of transition matrix in DTMC are positive, then the DTMC has to be [(d)].
- A DTMC is said to be [(e)] if all states communicate.
- A group of states that can reach each other is said to be a [(f)].
- In a DTMC, the limiting probability is same as the unique stationary distribution regardless of initial states if following three conditions of the DTMC are met: [(g)] [(h)] [(i)].
- State i is said to be [(j)] if, starting from state i , the probability of getting back to state i in some future is equal to 1.

Grading scheme:

- 2pts for each problems.
- 1pts for answering "communicate" in (f).
- 1pts for answering "absorbing" in (j).

#2. A small bank is staffed by a single server. It has been observed that, during a normal business day, the inter-arrival times of customers to the bank are iid having exponential distribution with mean 3 minutes. Also, the processing times of customers are iid having the following distribution (in minutes):

$$P(X = 1) = \frac{1}{4}, \quad P(X = 2) = \frac{1}{2}, \quad P(X = 3) = \frac{1}{4}.$$

An arrival finding the server busy joins the queue. The waiting space is infinite.

(a) What is long-run average waiting time of each customer in the queue? [10pts]

(b) What is long-run average number of customers in the bank? [10pts]

(a)

$$\mathbb{E}[U] = 3 \text{ min}, \quad \lambda = \frac{1}{3}, \quad c_a^2 = 1.$$

$$\mathbb{E}[V] = 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{2} + 3 \cdot \frac{1}{4} = \frac{1+4+3}{4} = 2 \text{ min}, \quad \mu = \frac{1}{2}.$$

$$\mathbb{E}[V^2] = 1^2 \cdot \frac{1}{4} + 2^2 \cdot \frac{1}{2} + 3^2 \cdot \frac{1}{4} = \frac{1+8+9}{4} = \frac{9}{2}.$$

$$\text{Var}(V) = \mathbb{E}[V^2] - (\mathbb{E}[V])^2 = \frac{9}{2} - 4 = \frac{1}{2},$$

$$c_s^2 = \frac{(\text{sd}(V))^2}{(\mathbb{E}[V])^2} = \frac{\frac{1}{2}}{4} = \frac{1}{8}.$$

$$\rho = \frac{\lambda}{\mu} = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}.$$

$$\mathbb{E}[W_q] = \mathbb{E}[V] \cdot \frac{\rho}{1-\rho} \cdot \frac{c_a^2 + c_s^2}{2} = 2 \cdot \frac{\frac{2}{3}}{1-\frac{2}{3}} \cdot \frac{1+\frac{1}{8}}{2} = \frac{9}{4} \text{ min.}$$

(b)

$$L_q = \lambda \mathbb{E}[W_q] = \frac{1}{3} \cdot \frac{9}{4} = \frac{3}{4},$$

$$L_s = \lambda \mathbb{E}[V] = \frac{1}{3} \cdot 2 = \frac{2}{3},$$

$$L_{\text{bank}} = L_q + L_s = \frac{3}{4} + \frac{2}{3} = \frac{17}{12} \approx 1.42.$$

Partial Grading: -5pts for each major error and -2.5pts for each minor error. Major/minor errors are determined by professor's discretion.

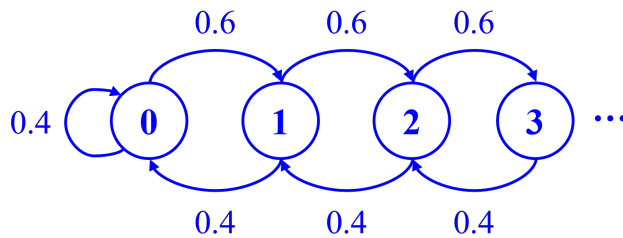
#3. This problem concerns whether the following statement is true or false.

If a DTMC is aperiodic and irreducible, then there exists a stationary distribution.

(a) Circle one of the following: [5pts] (**True/False**)

(b) If you answered "true", then explain why it is true and provide an example. If you answered "false", then provide a counterexample and explain it. [10pts]

- a) False
- b) The counter example must be provided for a periodic, irreducible, and infinite-state DTMC where stationary distribution does not exist. Infinite states random walk where all states are transient or null recurrent work. The following is a typical example:
Assume the random walk situation with $S = \{0, 1, 2, 3, \dots\}$.



This DTMC is aperiodic, irreducible and has infinite state space.

State	Inflow	Outflow	Relationship
State 0:	$0.4\pi_0 + 0.4\pi_1$	$0.6\pi_0 + 0.4\pi_0$	$\pi_1 = 3/2\pi_0$
State 1:	$0.4\pi_1 + 0.4\pi_2$	$0.6\pi_1 + 0.4\pi_1$	$\pi_2 = 3/2\pi_1$
State 2:	$0.4\pi_2 + 0.4\pi_3$	$0.6\pi_2 + 0.4\pi_2$	$\pi_3 = 3/2\pi_2$
State 3:	$0.4\pi_3 + 0.4\pi_4$	$0.6\pi_3 + 0.4\pi_3$	$\pi_4 = 3/2\pi_3$

The value of π_i is greater than 1, so it does not converge, and thus π_0 cannot be defined. Therefore, the stationary distribution cannot be defined.

Grade Scheme:

- (a) No partial points
- (b) If answered "true" in (a), then 0 pt. If very close efforts are made, then 5 pt.

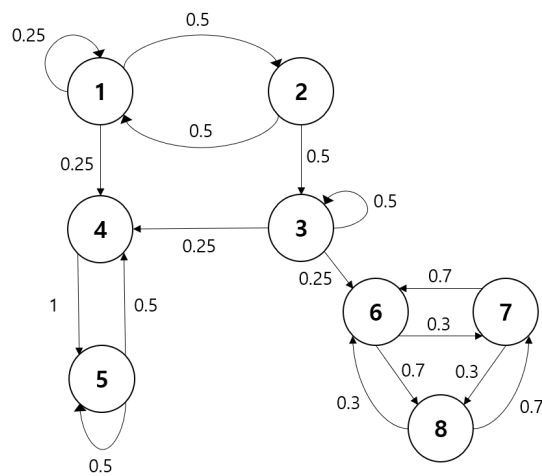
#4. Consider a DTMC with the following transition matrix.

$$\mathbf{P} = \begin{pmatrix} .25 & .5 & & .25 & & & & \\ .5 & & .5 & & & & & \\ & .5 & .25 & & .25 & & & \\ & & & 1 & & & & \\ & & & .5 & .5 & & & \\ & & & & & .3 & .7 & \\ & & & & & .7 & .3 & \\ & & & & & .3 & .7 & \end{pmatrix}$$

- (a) How many classes does it have? [5pts]
 (b) Calculate $\mathbf{P}^{100}$ (Show your work with proper justification.) [10pts]

Solution:

- (a) 4 classes.
 Recurrent classes: $\{4, 5\}$, $\{6, 7, 8\}$. Transient classes: $\{1, 2\}$, $\{3\}$.
- (b)
Step 1) Draw a transition diagram.



Step 2) Set up p^{100} and identify as many zeros as possible.

$$\mathbf{P}^{100} \approx \lim_{n \rightarrow \infty} \mathbf{P}^n = \begin{pmatrix} 0 & 0 & 0 & & & & & & \\ 0 & 0 & 0 & & & & & & \\ 0 & 0 & 0 & & & & & & \\ 0 & 0 & 0 & x & x & 0 & 0 & 0 & \\ 0 & 0 & 0 & x & x & 0 & 0 & 0 & \\ 0 & 0 & 0 & 0 & 0 & x & x & x & \\ 0 & 0 & 0 & 0 & 0 & x & x & x & \\ 0 & 0 & 0 & 0 & 0 & x & x & x & \end{pmatrix}$$

, where x is non zero number.

Step 3) Identify the “ x ” in above matrix. The “ x ” are limiting probability from a recurrent class to the recurrent class itself. Thus, we can find limiting probability by getting stationary distribution of the small Markov chain of recurrent class.

For the first recurrent class of $\{4, 5\}$,

$$\begin{aligned} \pi_4 &= 0.5\pi_5 \\ \pi_5 &= 0.1\pi_4 + 0.5\pi_5 \\ \pi_4 + \pi_5 &= 1 \end{aligned}$$

Thus, $(\pi_4, \pi_5) = (1/3, 2/3)$. Similarly, we have $(\pi_6, \pi_7, \pi_8) = (1/3, 1/3, 1/3)$. Now you have identified the “ x ”s in the previous matrix arriving to

$$\mathbf{P}^{100} \approx \lim_{n \rightarrow \infty} \mathbf{P}^n = \begin{pmatrix} 0 & 0 & 0 & & & & & & \\ 0 & 0 & 0 & & & & & & \\ 0 & 0 & 0 & & & & & & \\ 0 & 0 & 0 & 1/3 & 2/3 & 0 & 0 & 0 & \\ 0 & 0 & 0 & 1/3 & 2/3 & 0 & 0 & 0 & \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 1/3 & \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 1/3 & \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 1/3 & \end{pmatrix}$$

Step 4) We shall find the limiting probability going from transient state to recurrent class, i.e.

$f_{1,\{4,5\}}, f_{2,\{4,5\}}, f_{3,\{4,5\}}$.

Once you find these three numbers out, you will automatically get $f_{1,\{6,7,8\}}, f_{2,\{6,7,8\}}, f_{3,\{6,7,8\}}$.

$$\mathbf{P}^{100} \approx \lim_{n \rightarrow \infty} \mathbf{P}^n = \begin{pmatrix} 0 & 0 & 0 & 1/3 \cdot f_{1,\{4,5\}} & 2/3 \cdot f_{1,\{4,5\}} & 1/3 \cdot f_{1,\{6,7,8\}} & 1/3 \cdot f_{1,\{6,7,8\}} & 1/3 \cdot f_{1,\{6,7,8\}} \\ 0 & 0 & 0 & 1/3 \cdot f_{2,\{4,5\}} & 2/3 \cdot f_{2,\{4,5\}} & 1/3 \cdot f_{2,\{6,7,8\}} & 1/3 \cdot f_{2,\{6,7,8\}} & 1/3 \cdot f_{2,\{6,7,8\}} \\ 0 & 0 & 0 & 1/3 \cdot f_{3,\{4,5\}} & 2/3 \cdot f_{3,\{4,5\}} & 1/3 \cdot f_{3,\{6,7,8\}} & 1/3 \cdot f_{3,\{6,7,8\}} & 1/3 \cdot f_{3,\{6,7,8\}} \\ 0 & 0 & 0 & 1/3 & 2/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 2/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \end{pmatrix}$$

It remains us to find $f_{1,\{4,5\}}, f_{2,\{4,5\}}, f_{3,\{4,5\}}$ as explained above. So let’s begin filling this out by calculating the absorption probabilities into class $\{4, 5\}$

$$f_{1,\{4,5\}} = 0.25f_{1,\{4,5\}} + 0.5f_{2,\{4,5\}} + 0.25 \quad (1)$$

$$f_{2,\{4,5\}} = 0.5f_{1,\{4,5\}} + 0.5f_{3,\{4,5\}} \quad (2)$$

$$f_{3,\{4,5\}} = 0.5f_{3,\{4,5\}} + 0.25 \quad (3)$$

Solving by substituting the result of equation (3) (i.e. $f_{3,\{4,5\}} = 0.5$), in the equation (1) and (2) and solve a system of equations of them. we get

$$f_{1,\{4,5\}} = 3/4$$

$$f_{2,\{4,5\}} = 5/8$$

$$f_{3,\{4,5\}} = 1/2$$

Obviously, the absorption probabilities into class $\{6, 7, 8\}$ can be calculated directly since for the transient states the sum of all absorption probabilities is 1.

$$f_{1,\{6,7,8\}} = 1 - f_{1,\{4,5\}} = 1/4$$

$$f_{2,\{6,7,8\}} = 1 - f_{2,\{4,5\}} = 3/8$$

$$f_{3,\{6,7,8\}} = 1 - f_{3,\{4,5\}} = 1/2$$

Step 5) Deliver solution.

$$\mathbf{P}^{100} \approx \lim_{n \rightarrow \infty} \mathbf{P}^n = \begin{pmatrix} 0 & 0 & 0 & 1/4 & 1/2 & 1/12 & 1/12 & 1/12 \\ 0 & 0 & 0 & 5/24 & 5/12 & 1/8 & 1/8 & 1/8 \\ 0 & 0 & 0 & 1/6 & 1/3 & 1/6 & 1/6 & 1/6 \\ 0 & 0 & 0 & 1/3 & 2/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 2/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \end{pmatrix}$$

Grading scheme:

- (a) No partial points.
- (b) The partial points are given as follows:
 - 2.5pts : Step 1 - Step 2
 - 2.5pts : Step 3
 - 2.5pts : Step 4
 - 2.5pts : Step 5